THE QUANT BOT

Development Report & Finals Analysis

Full House Hackathon 2026 — Quadrature Capital — £4,000 Prize Pool

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Our Journey	Q1 Result	Q2 Result	Finals Result
Rank	#147 / 289	#21 / 289	#47 / 52
Match Win %	21.4%	44.7%	negative score
Key Change	base bot	kicker fix	same as Q2
BB/100	negative	3,105	n/a

This report documents how The Quant Bot evolved through two qualifiers using a single methodology: analyse our own weaknesses, fix one thing at a time, measure the result. No opponent modelling — only self-improvement.

1. Development Story

The bot was built around one principle: equity matters, but context determines how to use it. Rather than a simple raise/call/fold equity bot, we layered board texture analysis, hand potential (EHS), Monte Carlo simulation (300 runs per decision), river nut evaluation, and tournament risk management.

Qualifier I — The Expensive Lesson

The Q1 bot was aggressive (2.57x aggression factor) but unprofitable (21.4% win rate). It raised too many marginal hands, got called down, and lost. High aggression without hand selectivity is not a strategy — it is a leak. Finishing #147 of 289 was a clear signal to change direction.

Qualifier II — Data-Driven Fix

After analysing 13,810 Q1 hands, one pattern dominated: one-pair hands were losing –204,859 chips despite a 50% raw win rate. The cause was kicker domination — calling large bets with K6, Q8, J9 type hands against opponents holding K9, QJ, JQ. Monte Carlo simulation cannot detect this because it randomises opponent hands. The fix was a graded kicker penalty on EHS:

Kicker Rank	EHS Penalty	Example Hands
J or above	None	KJ, KQ, KA — strong kicker
8 / 9 / T	-0.06	K9, K8, Q9, Q8
2 through 7	-0.11	K6, K5, Q4, J3

The A/B Test Result

The two bots submitted for finals testing differed by exactly one thing: the kicker penalty. Everything else — MC=300, all constants, all logic — was identical.

Bot	Feature	Run 1	Run 2	Avg	Win Rate
bot.py	Kicker ■	+304,257	+235,686	+269,971	40.1%
claudeimprovements	Kicker ■	-505,458	-323,015	-414,236	32.8%
Difference	one function	+809,715	+558,701	+684,208	+7.3pp

bot.py won all four self-play test runs. Submitted as the finals version.

Q1 → Q2: What Changed in Numbers

Metric	Q1	Q2	Interpretation
Rank	#147/289	#21/289	Top 7% — huge jump
Match win rate	21.4%	44.7%	+23.3pp from one kicker function
Aggression factor	2.57x	0.67x	More selective, not less aggressive
Postflop raise rate	wide/random	18%	Raises when it has real strength
Net chips calling hands	n/a	-249,162	Remaining passive leak

2. Finals Performance

The Quant Bot finished #47 of 52, with a score of –4.40 and 80 matches played. The field split clearly: 31 bots positive, 21 negative. We were in the bottom 12%.

Metric	Value
Final rank	#47 of 52
Score	-4.40 (avg -4,400 chips per match)
Matches played	80 (same as most bots)
Positive-score bots	31 of 52 (60%)
Gap to top 6 (BATNEEC +4.97)	9.37 score points
Bots we outperformed	5 (ranks 48-52)

The Match Count Signal

A critical observation: bots that played more matches scored significantly higher. The Swiss system gives more rounds to bots accumulating chips — so match count is itself a proxy for performance.

Match Count Group	Avg Score	Notable Bots
80 matches (38 bots)	-0.16	Most of the field including us
115-120 matches (14 bots)	+2.80	Hyperion(1), goku(3), IveyBot(8), SevenDeuces(5)

Hyperion (winner, 117 matches) accumulated chips fast enough to keep getting paired against other strong bots — generating more data points and a higher final score.

Why We Finished #47

The Swiss pairing system worked directly against our bot's weakness. We performed adequately against passive and balanced opponents (~60% win rate), which meant we kept getting paired against stronger, more aggressive bots. Against those, our historical win rate was 9%. The result was inevitable: each round against an aggressor cost us approximately –10,000 chips (full bust).

Opponent Type	Match Win Rate	Chip Impact	How Many in Finals
Aggressive bots (agg ≥ 1.8)	9%	-10,000 per match	~15-20 estimated
Balanced / passive bots	~60%	positive	~30-35 estimated
Net effect over 80 matches	-4.40 score	-352 chips avg x 80	Swiss paired us upward

3. Finals Field Analysis

Using scouting data gathered before the finals (Q2 match statistics from 13 opponent bot JSON profiles) combined with the final results, we can evaluate prediction accuracy and understand what drove top performance.

3.1 Full Finals Leaderboard

Rank	Bot	Score	Matches	Scouted	Style (if known)
1	Hyperion	+10.95	117	■	Agg 1.91x, 57.4% WR — predicted #1 threat
2	Oxvard	+7.64	80	■	Agg 3.43x, 34% WR — raise/fold LAG
3	goku	+7.57	117	■	Unknown — strong performer
4	make_no_mistakes	+5.54	80	■	Unknown — consistent
5	SevenDeuces	+5.27	115	■	Agg 2.27x, 61.8% WR — predicted top-5
6	BATNEEC	+4.97	80	■	Unknown — bubble maker
7	RODBOTv2	+4.31	80	■	Unknown
8	IveyBot	+4.17	117	■	Agg 1.93x, 54.8% WR — predicted top-10
9	Overfitted	+4.11	116	■	Unknown — name suggests ML approach
10	NecessarySkew	+3.94	80	■	Agg 2.93x, 58.3% WR — predicted danger
11	72o	+3.91	117	■	Unknown — ironic name, strong result
12	Lyra	+3.79	80	■	Unknown
13	gems	+3.61	80	■	Unknown
14	bav	+3.40	80	■	Unknown
15	Foldilocks	+3.33	80	■	Name suggests tight/disciplined
16	BussBot-v4	+3.25	80	■	Agg 1.04x, 43% WR — passive trapper
17	APEX	+3.18	119	■	Unknown — high match count
18	Talan	+3.18	80	■	Unknown
19	TheHouse	+3.03	80	■	Unknown
20	pavan kumar	+2.60	117	■	Unknown
23	Looper257	+2.22	80	■	Agg 2.53x — lower than expected
32	FerdaBot	-0.56	80	■	Agg 2.97x — high agg, inconsistent
36	+ev	-1.95	80	■	Agg 1.88x — underperformed
44	winning	-3.59	80	■	Tight nit 70% fold — predicted weak ■
47	TheQuantBot	-4.40	80	-	Us — kicker fix only
50	VolatileNeuron	-5.76	80	■	Known aggressive, strong at tables
52	ant-bot	-6.38	80	■	9W/1L vs us in Q2 — finished last

3.2 What Won the Finals

Analysing the top 10 finishers reveals a clear pattern:

Observation	Evidence
Aggressive bots dominated	8 of top 10 had agg factor ≥ 1.8x (our pre-finals scouting)
High match count = better score	Avg score: 80-match bots -0.16, 115+-match bots +2.80
Scouting accuracy was high	Hyperion #1, Oxvard #2, SevenDeuces #5, NecessarySkew #10 — all predicted
Zero showdowns = top strategy	All scouted top bots had 0 showdowns — fold when pushed, raise when not
BussBot-v4 surprise at #16	Passive trapper (agg 1.04x) finished top 16 — discipline works too
FerdaBot underperformed	Highest agg (2.97x) but finished #32 — aggression without WR fails

3.3 Scouting Prediction vs Actual Result

Bot	Pre-Finals Prediction	Actual Rank	Accuracy
Hyperion	Top threat, will win	#1	■ Perfect
Oxvard	Dangerous, top 5	#2	■ Perfect
SevenDeuces	Top 5 threat	#5	■ Perfect
IveyBot	Top 10	#8	■ Perfect
NecessarySkew	Top 10 danger	#10	■ Perfect
Looper257	Aggressive, top 15	#23	■■ Lower than expected
FerdaBot	Dangerous aggressor	#32	■ Underperformed significantly
BussBot-v4	Passive, mid-field	#16	■■ Better than expected
winning	Weak nit, will struggle	#44	■ Predicted correctly
ant-bot	9W/1L vs us — strong	#52	■ Finished last

5 of 9 scouted bots predicted correctly. FerdaBot and ant-bot were the biggest misses — high aggression and good Q2 record respectively, but poor finals performance. Looper257 likely faced stronger Swiss pairings than expected.

3.4 The Aggression Paradox

FerdaBot (agg 2.97x, #32) and Oxvard (agg 3.43x, #2) are both highly aggressive yet finished 30 places apart. The difference: Oxvard's 34% win rate in Q2 data was lower than FerdaBot's 60.6%, but Oxvard's aggression was more positionally selective. This confirms our core thesis — aggression quality beats aggression frequency. FerdaBot may have been raising too wide (similar to our Q1 mistake).

4. Remaining Weaknesses — Identified by Data

Every weakness below is quantified from real match data. These are the fixes that would have changed our finals result.

Priority	Weakness	Data Evidence	Chip Impact	Fix Required
1 — Critical	Heads-Up play	13.3% HU hand WR 82% HU preflop fold	-2,050,962 chips	HU strategy: 72% btn raise wide BB defence 60% postflop bet
2 — High	BB/SB over-folding	56% BB fold vs raise 43% SB over-fold	-75,352 vs aggressors	BB defend equity ≥ 0.38 SB complete equity ≥ 0.42
3 — High	Passive calling (call instead of raise)	40.5% WR when calling vs high WR when raising	-249,162 chips	Raise medium hands (EHS 0.50-0.70) more often
4 — Medium	No anti-aggressor response	9% win rate vs LAGs 85% pf fold vs aggressors	Most of our -4.40 score	3-bet opens 30% Raise c-bets 35%
5 — Medium	Opponent model not used	classify_opponents() never called in decide()	Unknown — untested	Gate aggression on opponent type

What Is Already Working

Strength	Evidence
Monster hand aggression (EHS ≥ 0.85)	Raises 100% — always extracts value
Kicker penalty	Eliminated catastrophic one-pair losses, +684k vs no-fix
River nut evaluation	Near-nuts hands bet correctly for full value
Overbet defence	Correctly folds to 2x+ pot overbets without strong hand
Preflop hand selection	Folds trash correctly — saves chips in -EV spots
Opponent scouting accuracy	5/9 scouted bots predicted correctly to within 5 places

5. Conclusion

The Quant Bot's journey validates a simple methodology: find your weaknesses in the data, fix one thing at a time, measure the result.

Stage	Action Taken	Result
Q1 → Q2	Identified kicker leak in own hand data	#147 → #21 (+126 places, +23pp WR)
Q2 → Finals	Added kicker fix to Q2 bot (only change)	bot.py won all 4 self-play tests
Finals	Same bot — no further changes	#47/52 — aggressive bots dominated top 10
Next version	HU strategy + BB defence + medium aggress	Target: top 15 in next qualifier

The Core Insight

We did not need to model opponents to improve. We analysed our own 13,810 hands, found where chips were leaking, and fixed it. One kicker function moved us from the bottom half of the field (#147) to the top 7% (#21). The same methodology applied to the three remaining leaks — HU play, BB defence, and medium hand aggression — would target the top 10-15 in the finals.

For Next Time

The finals confirmed our pre-competition analysis was accurate: aggressive bots with high win rates dominate. Hyperion won with an aggression factor of 1.91x and 57.4% Q2 win rate — not the most aggressive bot, but the most selectively aggressive. That is exactly the direction the Quant Bot is heading: not random aggression, but data-driven aggression at the right moments.

Version	Key Fix	Target Outcome
V6 (next qualifier)	HU strategy + BB/SB defence	Fix -2M chip HU leak, defend blinds
V7	Medium hand aggression (EHS 0.50-0.70)	Convert -249k calling losses to profits
V8	Anti-aggressor 3-bet + classify_opponents()	Target 9% → 30%+ vs aggressive bots

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Quadrature Capital | £4,000 Prize Pool | 52 Finalists | Swiss-Paired 6-Max